

The Wharton School
Venture Capital and Entrepreneurial Management

Caselette #1: Organizational Issues in the Formation of a Start-Up

In 2005, two entrepreneurs, Tai and Chien-yu, want to form a new company in Santa Clara, California, to develop a systems-based optical switch which, when complete, will provide substantial additional bandwidth and speed to the fiber optics telecommunications industry. Chien-yu's strength is in engineering and product development and has been responsible for the development of the core product. Although Tai has been supportive during this developmental phase, his real expertise is in marketing—his efforts on behalf of their enterprise have focused to date on organizational and administrative matters, in anticipation of the new company's customer acquisition activities. Tai has also spent considerable time and effort in putting together a 50-page business plan for the company, which summarizes the company's markets, the technology and product focus of the new company, the marketing and market penetration strategy, an assessment of the potential and existing competitors of the company, a 3-year financial projection of anticipated results, and the bios of the two founders.

Although both of the founders have worked at big companies (Intel and Cisco) in the past, they have had no experience in either organizing or managing early stage companies, and no experience in seeking capital from outside investors. Chien-yu was a project manager at Intel responsible for a product development group of 15 engineers; Tai was a director-level employee while at Cisco, but had only spent a year at Cisco with even this level of responsibility before deciding to join up with Chien-yu to start their own company.

In the 24 months leading up to the launch of the new company, Chien-yu's effort to develop the core technology was supported with the assistance of a small chip design firm located in Portland, Oregon. Over the course of the two years that the development effort has been underway, approximately 10 engineers within this design firm have been informally engaged by Chien-yu to provide support on specific projects in bringing the prototype optical switch to completion. Chien-yu has periodically visited the design lab, but for the most part, communications have been handled remotely, and there have been periods of frustration resulting from inefficiencies and misstarts as a result of unclear specifications and failure of coordination. The services from the engineering group in Portland have been paid for by Chien-yu using his personal funds.

Chien-yu is also a part-time adjunct professor in the Computer Engineering Department at the University of California at Berkeley, where he teaches a seminar to a handful of Ph.D. students one quarter out of the year. Chien-yu occasionally uses his desktop computer and some of the specialized equipment in his laboratory on campus in running some key algorithms for the product design that he is developing, but is careful to maintain all of this personal data in a separate file on his computer while at work.

Chien-yu is willing to assign the complete rights to his technology over to the new company that he and Tai are forming. However, he believes that the core technology that will be the foundation of the optical switch system also may have potential commercial applications in the telecom area that are wholly outside the charter of the new company. Chien-yu has no intention of pursuing these alternative applications through the new company, in that the commercial markets for these alternative uses are, in his estimate, still years away. Nevertheless, Chien-yu is realistic about the chances for success of any new startup company, and as part of his long-range planning has included the possibility that the new company might fail and have to liquidate or sell out to a buyer in a

distress scenario. In an event such as this, Chien-yu wants to make sure he gets the technology assigned back to him, on the premise that he can retain the option to pursue these other commercial avenues.

Tai and Chien-yu believe that there is an enormous market in the telecommunications sector for their systems-based switch and are going to seek venture financing for their company. They believe they will immediately require approximately \$10 million in initial capital. With this money, they intend to:

- make substantial headway on the necessary chip design and overall product development. They believe they can demonstrate proof of concept with a major telecom partner in 18 months from the date of funding, with first commercial release of the product to take occur in the middle of their third year of operation.
- immediately begin hiring product engineers and marketing personnel as part of an ultimate headcount ramp that contemplates hiring 40 employees over the next 18 months. This will include discreet offers of employment to 5 of the engineers from the chip design firm that assisted in the development of the optical switch, since these engineers are familiar with the product and can immediately transition into the new company;
- establish the company's main office in Santa Clara, California as well as two adjunct locations in other geographic locations, including Portland, Oregon and Reston, Virginia. This will permit the company to recruit the best engineering resources on a nationwide basis without imposing the hardship of moving to the company's headquarter office.
- employ an executive search firm to locate an experienced Vice President of Strategic Development whose main responsibility would be business development with the major telecom companies; and
- provide Tai with a marketing budget to establish the company's presence in its customer markets in anticipation of product release in year 3.

Chien-yu and Tai have constructed a budget showing that the initial \$10 million in funding will last approximately 6 to 9 months, at which time the company will be in need of additional capital. Their business plan calls for all-in total financing of \$45 million, not an unusual amount of capital for a switch start-up, before the company can become profitable and generate sufficient cash to support its growth requirements. The founders don't expect to achieve profitability until the second half of the fourth year from the date of first funding.

Because of the significant amount of money they expect to lose during the initial stages of the company, the founders want to see if there is a business structure that will enable them to get some tax benefit for the projected losses.

Tai and Chien-yu want to obtain venture financing for their new company. They think their company should get a pre-money valuation of \$20 million. Although there are a number of private companies in the optical switch space that are competing for institutional venture capital funds, Tai and Chien-yu believe that all the market studies they have seen, combined with their systems-based product strategy, will assure them of a market penetration rate of at least 3% in the first 12 months following first commercial release of the product. Studies indicate that the total size of their market is at least \$4 billion. Ultimately, if the company is successful in executing its business plan, they would like to achieve an exit for the investors through an IPO of the company's stock.

The two founders have agreed that Chien-yu will be the President and Chief Executive Officer of the new company, and Tai will be the company's Executive Vice President of Marketing. Each will have an initial 50% equity interest in the startup company.

Questions:

You are a Wharton graduate and have recently joined a top-tier venture capital firm in Silicon Valley that has key interests in the telecom sector. One of the managing directors of the firm drops on your desk the 50-page business plan for Chien-yu's and Tai's new company as your first assignment and asks you to look into a number of issues relating to the plan, as follows:

1. Should the firm consider an investment in the new company? State your opinion and support it based on information from the proposal.
2. Chien-yu offers a reasonable case for personally recovering the intellectual property used in the company's optical switch product in the narrow case where the company is not successful. Do you agree or disagree? How would you respond to Chien-yu?
3. How should Tai and Chien-yu pay for the founders' stock that they will receive in the formation of the company?
4. If the venture firm decides to go forward with an investment, do you have any thoughts as to what the pre-money valuation of the new company should be? What are a few factors that you think should be addressed in determining the pre-money valuation? What percentage of the company will Tai and Chien-yu own if they are successful in negotiating a \$20 million pre-money valuation? Is this a reasonable equity stake for the founders to retain in the company upon completion of the first round of financing?
5. What are other strategies the founders could use in building the valuation in connection with the capital raising function? Is angel funding a likely financing resource for a company like this? What about strategic funding from a corporate investor such as TSMC or UMC?
6. What form of business entity could Tai and Chien-yu use to preserve the tax benefit associated with their expected losses?